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WATER RELEASE APPARATUS

Abstract

A water release apparatus includes a container having a cold-water valve mounting hole, a hot-water valve mounting hole, and a water release valve mounting hole; and a valve assembly including a cold-water valve, a hot-water valve, a water release valve respectively including a cold-water valve body, a hot-water valve body, and a water release valve body respectively seated in the cold-water valve mounting hole, the hot-water valve mounting hole, and the water release valve mounting hole and connected by a connection pipe. A cold-water valve bottom supporting member and a cold-water valve top supporting ring are mounted to the cold-water valve body. A hot-water valve bottom supporting member and a hot-water valve top supporting ring are mounted to the cold-water valve body. A water release valve bottom supporting member and a water release valve top supporting ring are mounted to a water release valve body.

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Background/Summary

BACKGROUND OF THE INVENTION

Technical Field of the Invention

[0001] The present invention relates to a water release apparatus, and more particularly to a water release apparatus that enables easy installation and part replacement.

Description of the Prior Art

[0002] Water release apparatuses, particularly water release apparatuses installed in residence houses, generally comprise a container (such as a wash basin, a bath tub, a kitchen sink, and the likes) and a valve assembly mounted on the container. However, actually, due to the difficulty of the operation of mounting the valve assembly and the container, it is common to separately purchase the valve assembly and the container and then having a plumber to install and set up the valve assembly and the container at home. This results in necessity of spending extra money to have plumbers to do the installation, in addition to the cost of purchases.

[0003] Further, the valve assembly, as being purchased, includes a valve contained therein and external decorations that are already set in a combined condition, making it hard to disassemble. Consequently, after a period of time of use by ordinary people, if one of the decorations is broken, it needs to replace the entire set of the valve assembly, and it is not possible to simply replace the broken part. This is a concern of money squandering.

SUMMARY OF THE INVENTION

[0004] Thus, in order to overcome the problems of the prior art water release apparatus that assembling is difficult and replacement of parts is not easy, the present invention provides a water release apparatus, which comprises: a container, which comprises a water outflow basin that is recessed and extended from a top side to a bottom side and a surface panel arranged at one side of the water outflow basin, the surface panel comprising a cold-water valve mounting hole, a hot-water valve mounting hole, and a water release valve mounting hole; and a valve assembly, which comprises a cold-water valve, a hot-water valve, a water release valve, and a connection pipe, the cold-water valve comprising a cold-water valve body, a cold-water valve bottom supporting member, and a cold-water valve top supporting ring, the cold-water valve body comprising an outside circumferential surface that comprises a cold-water valve lower thread and a cold-water valve upper thread, the cold-water valve bottom supporting member comprising a nut, the cold-water valve top supporting ring comprising a nut, the hot-water valve comprising a hot-water valve body, a hot-water valve bottom supporting member, and a hot-water valve top supporting ring, the hot-water valve body comprising an outside circumferential surface that comprises a hot-water valve lower thread and a hot-water valve upper thread, the hot-water valve bottom supporting member comprising a nut, the hot-water valve top supporting ring comprising a nut, the water release valve being arranged between the cold-water valve and the hot-water valve, the water release valve comprising a water release valve body, a water release valve bottom supporting member, and a water release valve top supporting ring, the water release valve body comprising an outside circumferential surface that comprises a water release valve thread, the water release valve bottom supporting member comprising a nut, the water release valve top supporting ring comprising a nut, the connection pipe connecting the cold-water valve body, the hot-water valve body, and the water release valve body, wherein the cold-water valve bottom supporting member is screwed to the cold-water valve lower thread, and the hot-water valve bottom supporting member is screwed to the cold-water valve lower thread, and the water release valve bottom supporting member is screwed to the water release valve thread, and the cold-water valve body, the hot-water valve body, and the water release valve body are respectively inserted, from the bottom side to the top side, into the cold-water valve mounting hole, the hot-water valve mounting hole, and the water release valve mounting hole of the container, such that the cold-water valve bottom supporting member, the hot-water valve bottom supporting member, and the water release valve bottom supporting member are positioned against an undersurface of the surface panel, and then, the cold-

water valve top supporting ring is screwed to the cold-water valve upper thread, and the hot-water valve top supporting ring is screwed to the hot-water valve upper thread, and the water release valve top supporting ring is screwed to the water release valve thread, such that the cold-water valve top supporting ring, the hot-water valve top supporting ring, and the water release valve top supporting ring are located on and positioned against a top surface of the surface panel, while the connection pipe is located on an underside of the surface panel. In this way, an effect that a user is allowed to do assembling easily and replacement of parts can be achieved easily.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is an exploded view showing a first preferred embodiment of the present invention.
[0006] FIG. 2 is a perspective view showing the embodiment of FIG. 1 in an assembled form.
[0007] FIG. 3 is a cross-sectional view showing the embodiment of FIG. 1 in an assembled form.
[0008] FIG. 4 is a schematic view showing installation of the embodiment of FIG. 1.
[0009] FIG. 5 is an exploded view showing a second preferred embodiment of the present invention.
[0010] FIG. 6 is a perspective view showing the embodiment of FIG. 5 in an assembled form.
[0011] FIG. 7 is a cross-sectional view showing the embodiment of FIG. 5 in an assembled form.
[0012] FIG. 8 is an exploded view showing a third preferred embodiment of the present invention.
[0013] FIG. 9 is a perspective view showing the embodiment of FIG. 8 in an assembled form.
[0014] FIG. 10 is a cross-sectional view showing the embodiment of FIG. 8 in an assembled form.
[0015] FIG. 11 is a cross-sectional view of the embodiment shown in FIG. 8 in an assembled form.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0016] Referring to FIGS. 1-4, a water release apparatus **100** according to a first preferred embodiment of the present invention mainly comprises a container **10** and a valve assembly **20**.
[0017] Referring to FIG. 1, the container **10** comprises a water outflow basin **11** that is recessed and extended from a top side to a bottom side and a surface panel **12** arranged at one side of a top of the water outflow basin **11**. The surface panel **12** comprises a cold-water valve mounting hole **121**, a hot-water valve mounting hole **122**, and a water release valve mounting hole **123**.
[0018] Referring to FIGS. 1-3, the valve assembly **20** comprises a cold-water valve **21**, a hot-water valve **22**, a water release valve **23**, and a connection pipe **24**. The cold-water valve **21** comprises a cold-water valve body **211**, a cold-water valve core **212** arranged in an interior of the cold-water valve body **211**, a cold-water pipe **213** connected to a bottom end of the cold-water valve body **211**, a cold-water valve bottom supporting member **214**, and a cold-water valve top supporting ring **215**. The cold-water valve body **211** has an outside circumferential surface that comprises a cold-water valve lower thread **216** and a cold-water valve upper thread **217**. An outside diameter of the cold-water valve upper thread **217** is smaller than an outside diameter of the cold-water valve lower thread **216**, such that a cold-water valve stepped face **218** is formed on the outside surface of the cold-water valve body **211** between the cold-water valve lower thread **216** and the cold-water valve upper thread **217**. The cold-water valve bottom supporting member **214** is a nut, and the cold-water valve top supporting ring **215** is a nut. The hot-water valve **22** comprises a hot-water valve body **221**, a hot-water valve core **222** arranged in an interior of the hot-water valve body **221**, a hot-water pipe **223** connected to a bottom end of the hot-water valve body **221**, a hot-water valve bottom supporting member **224**, and a hot-water valve top supporting ring **225**. The hot-water valve body **221** has outside circumferential surface that comprises a hot-water valve lower thread **226** and a hot-water valve upper thread **227**. An outside diameter of the hot-water valve upper thread **227** is smaller than an outside diameter of the hot-water valve lower thread **226**, such that a hot-water valve stepped face **228** is formed on the outside surface of the hot-water valve body **221** between

the hot-water valve lower thread **226** and the hot-water valve upper thread **227**. The hot-water valve bottom supporting member **224** is a nut, and the hot-water valve top supporting ring **225** is a nut. The water release valve **23** is arranged between the cold-water valve **21** and the hot-water valve **22**, and the water release valve **23** comprises a water release valve body **231**, a water release valve bottom supporting member **232**, and a water release valve top supporting ring **233**. The water release valve body **231** has an outside circumferential surface that comprises a water release valve thread **234**. The water release valve bottom supporting member **232** is a nut, and the water release valve top supporting ring **233** is a nut. The connection pipe **24** connects the cold-water valve body **211** of the cold-water valve **21**, the hot-water valve body **221** of the hot-water valve **22**, and the water release valve body **231** of the water release valve **23**.

[0019] Thus, the above provides a description of various components of the water release apparatus **100** according to one preferred embodiment of the present invention, and in the following, assembly and features thereof will be described as follows.

[0020] To assemble the present invention, the cold-water valve bottom supporting member **214** is first screwed to the cold-water valve lower thread **216**, and the hot-water valve bottom supporting member **224** is screwed to the hot-water valve lower thread **226**, and the water release valve bottom supporting member **232** is screwed to the water release valve thread **234**. Then, the cold-water valve body **211**, the hot-water valve body **221**, and the water release valve body **231** are inserted, from the bottom side toward the top side, into the cold-water valve mounting hole **121**, the hot-water valve mounting hole **122**, and the water release valve mounting hole **123** of the container **10**, respectively (as shown in FIGS. 2 and 3), such that the cold-water valve bottom supporting member **214**, the hot-water valve bottom supporting member **224**, and the water release valve bottom supporting member **232** are positioned against an undersurface of the surface panel **12** of the container **10** (as shown in FIG. 3). And, then, the cold-water valve top supporting ring **215** is screwed to the cold-water valve upper thread **217**, and the hot-water valve top supporting ring **225** is screwed to the hot-water valve upper thread **227**, and the water release valve top supporting ring **233** is screwed to the water release valve thread **234**, such that the cold-water valve top supporting ring **215**, the hot-water valve top supporting ring **225**, and the water release valve top supporting ring **233** are located on and positioned against a top surface of the surface panel **12** of the container **10**, while the connection pipe **24** is located on the underside of the surface panel **12** (as shown in FIG. 3).

[0021] As such, the operation of assembling present invention can be carried out in advance in a workshop to be then shipped to a sales shop for exhibition or directly delivered to a purchaser's residence to thereby reduce the difficulty for the user to do installation by himself or herself.

[0022] Then, the user only needs to carry out easy mounting of a water dispense port **91**, switch knobs **92**, and other decorative panels accompanying therewith or additionally purchased (as shown in FIG. 4) to complete the entire operation of installation. In this way, the present invention provides an effect of easy replacement of components (the water dispense port, the switch knobs, and other decorative panels).

[0023] Further, since the cold-water valve **21** and the hot-water valve **22** are respectively provided with the cold-water valve stepped face **218** and the hot-water valve stepped face **228**, the positions of the cold-water valve top supporting ring **215** and the hot-water valve top supporting ring **225** achieved with downward movements thereof can be limited to thereby make heights of the cold-water valve body **211** and the hot-water valve body **221** projecting beyond the top surface of the surface panel **12** fixed to facilitate the user's operation of mounting the water dispense port **91**, the switch knobs **92**, and other decorative panels, thereby ensuring consistency for aesthetics.

[0024] Referring to FIGS. 5-7, a water release apparatus **200** according to a second preferred embodiment of the present invention mainly comprises a container **10** and a valve assembly **30**.

[0025] Referring to FIG. 5, the container **10** comprises a water outflow basin **11** that is recessed and extended from a top side to a bottom side and a surface panel **12** arranged at one side of a top

of the water outflow basin **11**. The surface panel **12** comprises a cold-water valve mounting hole **121**, a hot-water valve mounting hole **122**, and a water release valve mounting hole **123**.

[0026] Referring to FIGS. 5-7, the valve assembly **30**, comprises a cold-water valve **31**, a hot-water valve **32**, a water release valve **33** and a connection pipe **34**. The cold-water valve **31** comprises a cold-water valve body **311**, a cold-water valve core **312** arranged in an interior of the cold-water valve body **311**, a cold-water pipe **313** connected to a bottom end of the cold-water valve body **311**, a cold-water valve bottom supporting member **314**, and a cold-water valve top supporting ring **315**. The cold-water valve body **311** has an outside circumferential surface that comprises a cold-water valve thread **316**. The cold-water valve bottom supporting member **314** integrally projects outwards, in a radial direction, from the outside circumferential surface of the cold-water valve body **311**. The cold-water valve top supporting ring **315** is a nut. The hot-water valve **32** comprises a hot-water valve body **321**, a hot-water valve core **322** arranged in an interior of the hot-water valve body **321**, a hot-water pipe **323** connected to a bottom end of the hot-water valve body **321**, a hot-water valve bottom supporting member **324**, and a hot-water valve top supporting ring **325**. The hot-water valve body **321** has an outside circumferential surface that comprises a hot-water valve thread **326**. The hot-water valve bottom supporting member **324** integrally projects outwards, in a radial direction, from the outside circumferential surface of the hot-water valve body **321**. The hot-water valve top supporting ring **325** is a nut. The water release valve **33** is arranged between the cold-water valve **31** and the hot-water valve **32**, and the water release valve **33** comprises a water release valve body **331** and a water release valve top supporting ring **332**. The water release valve body **331** has an outside circumferential surface that comprises a water release valve thread **333**. The water release valve top supporting ring **332** is a nut. The connection pipe **34** connects the cold-water valve body **311** of the cold-water valve **31**, the hot-water valve body **321** of the hot-water valve **32**, and the water release valve body **331** of the water release valve **33**.

[0027] To assemble the instant embodiment, the cold-water valve body **311**, the hot-water valve body **321**, and the water release valve body **33** are inserted, from the bottom side toward the top side, into the cold-water valve mounting hole **121**, the hot-water valve mounting hole **122**, and the water release valve mounting hole **123** of the container **10**, respectively (as shown in FIGS. 6 and 7), such that the cold-water valve bottom supporting member **314** and the hot-water valve bottom supporting member **324** are positioned against an undersurface of the surface panel **12** of the container **10** (as shown in FIG. 7). And, then, the cold-water valve top supporting ring **315** is screwed to the cold-water valve thread **316**, and the hot-water valve top supporting ring **325** is screwed to the hot-water valve thread **326**, and the water release valve top supporting ring **332** is screwed to the water release valve thread **333**, such that the cold-water valve top supporting ring **315**, the hot-water valve top supporting ring **325**, and the water release valve top supporting ring **332** are located on and positioned against a top surface of the surface panel **12** of the container **10**, while the connection pipe **34** is located on the underside of the surface panel **12** (as shown in FIG. 7).

[0028] Thus, the above provides a description of various components of the water release apparatus **200** according to the second embodiment of the present invention and the way of assembling thereof, which, although different from those of the first embodiment, yet achieve the same effect of easy assembling and convenient replacement of components.

[0029] Referring to FIGS. 8-11, a water release apparatus **300** according to a third preferred embodiment of the present invention mainly comprises a container **10** and a valve assembly **40**.

[0030] Referring to FIG. 8, the container **10** comprises a water outflow basin **11** that is recessed and extended from a top side to a bottom side and a surface panel **12** arranged at one side of a top of the water outflow basin **11**. The surface panel **12** comprises a cold-water valve mounting hole **121**, a hot-water valve mounting hole **122**, and a water release valve mounting hole **123**.

[0031] Referring to FIGS. 8-11, the valve assembly **40** comprises a cold-water valve **41**, a hot-water valve **42**, a water release valve **43**, and a connection pipe **44**. The cold-water valve **41**

comprises a cold-water valve body **411**, a cold-water valve core **412** arranged in an interior of the cold-water valve body **411**, a cold-water pipe **413** connected to a bottom end of the cold-water valve body **411**, a cold-water valve bottom supporting member **414**, and a cold-water valve top supporting ring **415**. The cold-water valve body **411** has an outside circumferential surface that comprises a cold-water valve thread **416**. The cold-water valve bottom supporting member **414** integrally projects outwards, in a radial direction, from the outside circumferential surface of the cold-water valve body **411**. The cold-water valve top supporting ring **415** is a nut. The hot-water valve **42** comprises a hot-water valve body **421**, a hot-water valve core **422** arranged in an interior of the hot-water valve body **421**, a hot-water pipe **423** connected to a bottom end of the hot-water valve body **421**, a hot-water valve bottom supporting member **424**, and a hot-water valve top supporting ring **425**. The hot-water valve body **421** has an outside circumferential surface that comprises a hot-water valve thread **426**. The hot-water valve bottom supporting member **424** integrally projects outwards, in a radial direction, from the outside circumferential surface of the hot-water valve body **421**. The hot-water valve top supporting ring **425** is a nut. The water release valve **43** is arranged between the cold-water valve **41** and the hot-water valve **42**, and the water release valve **43** comprises a water release valve body **431**, a water release valve bottom supporting member **432**, and a water release valve top supporting ring **433**. The water release valve body **431** has an outside circumferential surface that comprises a water release valve thread **434**. The water release valve bottom supporting member **432** integrally projects outwards, in a radial direction, from the outside circumferential surface of the water release valve body **431**. The water release valve top supporting ring **433** is a nut. The connection pipe **44** connects the cold-water valve body **411** of the cold-water valve **41**, the hot-water valve body **421** of the hot-water valve **42**, and the water release valve body **431** of the water release valve **43**.

[0032] To assemble the instant embodiment, the cold-water valve body **411**, the hot-water valve body **421**, and the water release valve body **431** are inserted, from the bottom side toward the top side, into the cold-water valve mounting hole **121**, the hot-water valve mounting hole **122**, and the water release valve mounting hole **123** of the container **10**, respectively (as shown in FIGS. **9** and **10**), such that the cold-water valve bottom supporting member **414**, the hot-water valve bottom supporting member **424**, and the water release valve bottom supporting member **432** are positioned against an undersurface of the surface panel **12** of the container **10** (as shown in FIG. **10**). And, then, the cold-water valve top supporting ring **415** is screwed to the cold-water valve thread **416**, and the hot-water valve top supporting ring **425** is screwed to the hot-water valve thread **426**, and the water release valve top supporting ring **433** is screwed to the water release valve thread **434**, such that the cold-water valve top supporting ring **415**, the hot-water valve top supporting ring **425**, and the water release valve top supporting ring **433** are located on and positioned against a top surface of the surface panel **12** of the container **10**, while the connection pipe **44** is located on the underside of the surface panel **12** (as shown in FIG. **10**).

[0033] Thus, the above provides a description of various components of the water release apparatus **300** according to the third embodiment of the present invention and the way of assembling thereof, which, although different from those of the previous embodiments, yet achieve the same effect of easy assembling and convenient replacement of components.

[0034] Further, although in the first embodiment and the second embodiment, the cold-water valve bottom supporting member, the hot-water valve bottom supporting member, and the water release valve bottom supporting member are made in the form of a ring, in the instant embodiment the cold-water valve bottom supporting member, the hot-water valve bottom supporting member, and the water release valve bottom supporting member are each made in the form of two tabs integrally project outwards, in a radial direction, from the outside circumferential surface of the cold-water valve body, the hot-water valve body, and the water release valve body. Although the shapes are different, the effect of supporting is the same.

[0035] Further, in the instant embodiment, as shown in FIG. **8**, the cold-water valve **41** further

comprises two cold-water valve screws **417**, and the two cold-water valve screws **417** respectively penetrate, from the bottom side to the top side, through the two tabs of the cold-water valve bottom supporting member **414** to then screw to and attach to the undersurface of the surface panel **12**. The hot-water valve **42** further comprises two hot-water valve screws **427**, and the two hot-water valve screws **427** respectively penetrate, from the bottom side to the top side, through the two tabs of the hot-water valve bottom supporting member **424** to then screw to and attach to the undersurface of the surface panel **12**. The water release valve **43** further comprises two water release valve screws **435**, and the two water release valve screws **435** respectively penetrate, from the bottom side to the top side, through the two tabs of the water release valve bottom supporting member **432** to then screw to and attach to the undersurface of the surface panel **12** (as shown in FIG. **11**). As such, the strength of combination between the valve assembly **40** and the container **10** is enhanced.

Claims

1. A water release apparatus, comprising: a container, which comprises a water outflow basin that is recessed and extended from a top side to a bottom side and a surface panel arranged at one side of the water outflow basin, the surface panel comprising a cold-water valve mounting hole, a hot-water valve mounting hole, and a water release valve mounting hole; and a valve assembly, which comprises a cold-water valve, a hot-water valve, a water release valve, and a connection pipe, the cold-water valve comprising a cold-water valve body, a cold-water valve bottom supporting member, and a cold-water valve top supporting ring, the cold-water valve body comprising an outside circumferential surface that comprises a cold-water valve lower thread and a cold-water valve upper thread, the cold-water valve bottom supporting member comprising a nut, the cold-water valve top supporting ring comprising a nut, the hot-water valve comprising a hot-water valve body, a hot-water valve bottom supporting member, and a hot-water valve top supporting ring, the hot-water valve body comprising an outside circumferential surface that comprises a hot-water valve lower thread and a hot-water valve upper thread, the hot-water valve bottom supporting member comprising a nut, the hot-water valve top supporting ring comprising a nut, the water release valve being arranged between the cold-water valve and the hot-water valve, the water release valve comprising a water release valve body, a water release valve bottom supporting member, and a water release valve top supporting ring, the water release valve body comprising an outside circumferential surface that comprises a water release valve thread, the water release valve bottom supporting member comprising a nut, the water release valve top supporting ring comprising a nut, the connection pipe connecting the cold-water valve body, the hot-water valve body, and the water release valve body; wherein the cold-water valve bottom supporting member is screwed to the cold-water valve lower thread, and the hot-water valve bottom supporting member is screwed to the hot-water valve lower thread, and the water release valve bottom supporting member is screwed to the water release valve thread; the cold-water valve body, the hot-water valve body, and the water release valve body are respectively inserted, from the bottom side toward the top side, into the cold-water valve mounting hole, the hot-water valve mounting hole, and the water release valve mounting hole of the container, such that the cold-water valve bottom supporting member, the hot-water valve bottom supporting member, and the water release valve bottom supporting member are positioned against an undersurface of the surface panel; then, the cold-water valve top supporting ring is screwed to the cold-water valve upper thread, and the hot-water valve top supporting ring is screwed to the hot-water valve upper thread, and the water release valve top supporting ring is screwed to the water release valve thread, such that the cold-water valve top supporting ring, the hot-water valve top supporting ring, and the water release valve top supporting ring are located on and positioned against a top surface of the surface panel, while the connection pipe is located on an underside of the surface panel.

2. The water release apparatus according to claim 1, wherein the cold-water valve further

comprises a cold-water valve core arranged in an interior of the cold-water valve body and a cold-water pipe connected to a bottom end of the cold-water valve body; the hot-water valve further comprises a hot-water valve core arranged in an interior of the hot-water valve body and a hot-water pipe connected to a bottom end of the hot-water valve body.

3. The water release apparatus according to claim 1, wherein an outside diameter of the cold-water valve upper thread is smaller than an outside diameter of the cold-water valve lower thread, such that a cold-water valve stepped face is formed on the outside surface of the cold-water valve body between the cold-water valve lower thread and the cold-water valve upper thread; an outside diameter of the hot-water valve upper thread is smaller than an outside diameter of the hot-water valve lower thread, such that a hot-water valve stepped face is formed on the outside surface of the hot-water valve body between the hot-water valve lower thread and the hot-water valve upper thread.

4. A water release apparatus, comprising: a container, which comprises a water outflow basin that is recessed and extended from a top side to a bottom side and a surface panel arranged at one side of the water outflow basin, the surface panel comprising a cold-water valve mounting hole, a hot-water valve mounting hole, and a water release valve mounting hole; and a valve assembly, which comprises a cold-water valve, a hot-water valve, a water release valve, and a connection pipe, the cold-water valve comprising a cold-water valve body, a cold-water valve bottom supporting member, and a cold-water valve top supporting ring, the cold-water valve body comprising an outside circumferential surface that comprises a cold-water valve thread, the cold-water valve bottom supporting member integrally projecting outwards, in a radial direction, from the outside circumferential surface of the cold-water valve body, the cold-water valve top supporting ring comprising a nut, the hot-water valve comprising a hot-water valve body, a hot-water valve bottom supporting member, and a hot-water valve top supporting ring, the hot-water valve body comprising an outside circumferential surface that comprises a hot-water valve thread, the hot-water valve bottom supporting member integrally projecting outwards, in a radial direction, from the outside circumferential surface of the hot-water valve body, the hot-water valve top supporting ring comprising a nut, the water release valve being arranged between the cold-water valve and the hot-water valve, the water release valve comprising a water release valve body and a water release valve top supporting ring, the water release valve body comprising an outside circumferential surface that comprises a water release valve thread, the water release valve top supporting ring comprising a nut, the connection pipe connecting the cold-water valve body, the hot-water valve body, and the water release valve body; wherein the cold-water valve body, the hot-water valve body, and the water release valve body are respectively inserted, from the bottom side toward the top side, into the cold-water valve mounting hole, the hot-water valve mounting hole, and the water release valve mounting hole of the container, such that the cold-water valve bottom supporting member and the hot-water valve bottom supporting member are positioned against an undersurface of the surface panel; then, the cold-water valve top supporting ring is screwed to the cold-water valve thread, and the hot-water valve top supporting ring is screwed to the hot-water valve thread, and the water release valve top supporting ring is screwed to the water release valve thread, such that the cold-water valve top supporting ring, the hot-water valve top supporting ring, and the water release valve top supporting ring are located on and positioned against a top surface of the surface panel, while the connection pipe is located on an underside of the surface panel.

5. The water release apparatus according to claim 4, wherein the cold-water valve further comprises a cold-water valve core arranged in an interior of the cold-water valve body and a cold-water pipe connected to a bottom end of the cold-water valve body; the hot-water valve further comprises a hot-water valve core arranged in an interior of the hot-water valve body and a hot-water pipe connected to a bottom end of the hot-water valve body.

6. The water release apparatus according to claim 4, wherein the cold-water valve bottom supporting member and the hot-water valve bottom supporting member are each in the form of a

ring.

7. The water release apparatus according to claim 4, wherein the cold-water valve bottom supporting member and the hot-water valve bottom supporting member are each in the form of two tabs.

8. The water release apparatus according to claim 4, wherein the water release valve further comprises a water release valve bottom supporting member, and the water release valve bottom supporting member integrally projects outwards, in a radial direction, from the outside circumferential surface of the water release valve body.

9. The water release apparatus according to claim 8, wherein the water release valve bottom supporting member comprises two tabs.

10. The water release apparatus according to claim 8, wherein the cold-water valve further comprises at least one cold-water valve screw, the cold-water valve screw penetrating, from the bottom side to the top side, through the cold-water valve bottom supporting member to screw to and attach to the surface panel; the hot-water valve further comprises at least one hot-water valve screw, the hot-water valve screw penetrating, from the bottom side to the top side, through the hot-water valve bottom supporting member to screw to and attach to the surface panel; the water release valve further comprises at least one water release valve screw, the water release valve screw penetrating, from the bottom side to the top side, through the water release valve bottom supporting member to screw to and attach to the surface panel.
